

NEET Important Chapter – Nuclei

Nuclei An Important Concept of NEET

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Nuclear physics is the branch of physics which studies the nature of nuclei. The atoms are said to be revolving around nuclei. The nucleus of an atom consists of neutrons and protons, collectively called nucleons.

This chapter is a part of Modern Physics and considered very important for the NEET exam. In this chapter we will study about the important topics of Nuclei such as properties of nucleus and radioactivity.

In this article, we will be focusing on the concepts of Nuclei that play a vital role in the NEET exam.

Important Topics of Nuclei

- Nuclear size and nuclear density
- Mass energy relation and nuclear binding energy
- Radioactivity
- Alpha, beta and Gamma decay
- Nuclear fission and fusion

Important Concepts of Nuclei

We know that in a chapter there will be many concepts that are being mentioned in the textbook, but when it comes to NEET preparation we have to focus on what is most important among all the concepts in the chapter. Below is the table provided for some most important concepts of Nuclei.

Sl. No	Name of the Concept	Key Points of Concept
1.	Nuclear size and nuclear density	- The radius of the nucleus is given by $R = R_0 A^{1/3}$. Where, $R_0 = 1.3$ fermi and $A =$ mass number. - Volume of Nucleus is given by $V = \frac{4}{3}\pi R^3$
2.	Mass energy relation and nuclear binding energy	- Mass energy relation tells us that when a certain mass (Δm) disappears then an equivalent amount of energy (ΔE) appears and vice-versa. - The energy with which nucleons in a nucleus are bonded is referred to as the nucleus' binding energy. It is determined by the amount of effort necessary to detach the nucleons from the nucleus at an infinite distance.

3.	Radioactivity	• Radioactivity is the process by virtue of which a heavy element disintegrates itself without being forced by any external agent to do so. • According to laws of radioactive disintegration, the number of atoms disintegrating per second at any instant is directly proportional to the number of radioactive atoms actually present in the sample at that instant. • The half life of a radioactive element is the amount of time it takes for half of the atoms present in the sample to decay. • Average life time of the element is obtained by calculating the total life time of all the atoms of the element and dividing it by the total number of atoms present initially in the sample of the element

4.	Alpha, beta and Gamma decay	- Alpha decay is the phenomenon of emission of an alpha particle from a radioactive nucleus. Ex: - Beta decay is the phenomenon of emission of an electron from a radioactive nucleus. Ex: - Gamma decay is the phenomenon of emission of gamma ray photons from a radioactive nucleus. Ex:
5.	Nuclear fission and fusion	- Nuclear fission is the process of splitting a heavy nucleus (usually $A > 230$) into two or more lighter nuclei. - Nuclear fusion is the process by which two or more lighter nuclei combine to produce a single heavy nucleus.

List of Important Formulae of Nuclei

Sl. No	Name of Concept	Formula

1.	Nuclear size and density	The radius of the nucleus is given by $R = R_0 A^{1/3}$. Where, $R_0 = 1.3$ fermi and A = mass number. Volume of Nucleus is given by —
2.	Law of Radioactive Decay	Where, N = number of nuclei remaining at time, t N_0 = Number of nuclei at time, $t=0$ and λ = decay constant Half life ($T_{1/2}$) = — Mean/ Average life (T_{av}) = —

3.	Mass energy relation and nuclear binding energy	Mass energy relation is given as, Binding energy of the nucleus, Here, Z = charge number, A = mass number, m_p = mass of the proton, m_n = mass of the neutron and m = mass of nucleus.
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Solved Examples of Nuclei

1. The most abundant isotope of helium is the ${}^4_2\text{He}$ nucleus whose mass is 4.0026 kg. For this nucleus, find

- (a) the mass defect
- (b) the binding energy.

Sol:

The symbol ${}^4_2\text{He}$ indicates that the helium nucleus contains 2 protons and 2 neutrons ($N = 4 - 2 = 2$). To obtain mass defect Δm , we first determine the sum of masses of protons and neutrons. Then, we subtract this sum from the mass of the nucleus.

(a) Now, sum of masses of protons and neutrons = $2(\text{Mass of a proton}) + 2(\text{mass of a neutron})$

$$= 2(1.6726 \times 10^{-27} \text{ kg}) + 2(1.6749 \times 10^{-27} \text{ kg}) = 6.6950 \times 10^{-27} \text{ kg}$$

Now,

Mass defect (Δm) = Mass of nucleus - sum of masses of protons and neutrons

$$\Delta m = \quad \quad \quad \text{kg}$$

$$\Delta m = \quad \quad \quad \text{kg}$$

Therefore, the mass defect is $\Delta m = \quad \quad \quad \text{kg}$

(b) Now, binding energy is given by

$$\Delta E_{BE} = (\Delta m)c^2 = \quad \quad \quad \text{J}$$

Or we can write in eV = _____

So, binding energy in eV is 28.3 eV.

Key point: To obtain mass defect, always first find the sum of mass of protons and neutrons and subtract it by the mass of the nucleus.

2. The mean lives of an unstable nucleus in two different decay processes are 1620 yr and 405 yr, respectively. Find out the time during which three-fourth of a sample will decay.

Sol:

Given,

Mean life of Process- 1 (t_1) = 1620 yr,

Mean life of Process- 2 (t_2) = 405 yr,

Let decay constants of process-1 and process- 2 be λ_1 and λ_2 , respectively. Then,

—

—

If the effective decay constant is λ , then

$$\lambda N = \lambda_1 N + \lambda_2 N$$

$$\lambda = \lambda_1 + \lambda_2$$

$$\frac{1}{4} = \frac{1}{2} e^{-\lambda t} \quad \text{year}^{-1}$$

$$\frac{1}{4} = \frac{1}{2} e^{-\lambda t} \quad \text{year}^{-1}$$

Now, when three-fourth of the sample will decay. The sample remaining will be one-fourth of the total sample. If the initial sample was N_0 , then final sample will have $N_0/4$,

So, —

Applying logarithms on both the sides, we get,

$$\ln \frac{1}{4} = \ln \frac{1}{2} e^{-\lambda t} = -1.386$$

$$t = 449 \text{ yr}$$

Therefore, the time during which three-fourth of a sample will decay will be 449 yr.

Key point: We need to modify the formula according to the given data.

Previous Year Questions from Nuclei

1. The half life of a radioactive nuclide is 100 hrs. The fraction of original activity that will remain after 150 h would be: (NEET 2021)

1. —

2. —=

3. —

4. —=

Sol:

Given,

Half life ($t_{1/2}$) = 100 hrs,

At $t = 150$ hrs,

$A =$ —

Where, A = remaining activity of nuclide after time t and A_0 = original activity of nuclide,

— = — =

So, the fraction of activity that will remain after 150 h will be — or —=

Therefore, option- (b) is the right answer.

Trick: Most questions repeat on the basis of a similar formula.

2. There are — radioactive nuclei in a given radioactive element, its half-life time is 1 minute. How many nuclei will remain after 30 seconds? (Take, —) (NEET 2021)

a.

b.

c.

d.

Sol:

Given that,

Original number of nuclei,

Half- life time, $t_{1/2}$

To find: Number of nuclei remain after time ($t = 30$ seconds) i.e, .

To solve this problem we have to use the concept of half life and also the relation of the number of nuclei decayed to the original number of nuclei present in the sample of a radioactive element.

Now using the concept of half live, we can write the relation between the number of nuclei decayed () to the original number of nuclei() as,

$$\frac{N_0 - N}{N_0} = 1 - \left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}}$$

Now after putting the values of the quantities in the above relation, we get;

$$\frac{N_0 - N}{N_0} = 1 - \left(\frac{1}{2}\right)^{\frac{30}{15}}$$

$$\frac{N_0 - N}{N_0} = 1 - \left(\frac{1}{2}\right)^2$$

$$\frac{N_0 - N}{N_0} = \frac{3}{4}$$

Hence, the number of nuclei that remain after 30 seconds is .

Therefore, option a is correct.

Trick: The application of the half life concept and the relation between the number of nuclei decayed to the original number of nuclei in a sample is important to solve this problem.

Practice Questions

1. The half-lives of radioisotopes P_{32} and P_{33} are 14 days and 25 days respectively. These radioisotopes are mixed in the ratio of 4 : 1 of their atoms. If the initial activity of the mixed sample is 3.0 mCi, find the activity of the mixed isotopes after 60 years. (Ans: 0.205 mCi)
2. A stable nuclei C is formed from two radioactive nuclei A and B with decay constant of λ_1 and λ_2 respectively. Initially, the number of nuclei of A is N_0 and that of B is zero. Nuclei B are produced at a constant rate of P. Find the number of the nuclei of C after time t (Ans: _____)

Conclusion

In this article we have discussed the Nuclei chapter with respect to the NEET exam. In this article we have provided solved examples, previous year questions, important formulae list, etc..

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